# Dark-sector residuals in compact $S^3$ FLRW cosmology

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We formulate compact closed-FLRW cosmology on $\mathcal{M} = \mathbb{R} \times S^3$ as a conditional reconstruction problem for the gravitational dark residual. The assumptions are deliberately limited: general relativity, a closed homogeneous and isotropic metric, an independently specified visible sector, and a specified or reconstructed expansion history $E(a) = H(a)/H_0$. Compact $S^3$ geometry supplies finite volume, finite diameter, positive spatial curvature, causal accessibility bounds, compact covariance kernels, and a discrete scalar–vector–tensor harmonic bookkeeping system. It does not by itself imply dark matter, accelerated expansion, $w = -1$, or a cosmological constant. The dark sector is instead defined by the covariant tensor residual $T^{\mathcal{D}}_{\mu\nu} = c^4 G_{\mu\nu}/(8\pi G) - T^{\rm vis}_{\mu\nu}$. For a visible baryon–photon–neutrino sector, the closed-FLRW background fixes $\epsilon_{\mathcal{D}}(a)$ and $p_{\mathcal{D}}(a)$ uniquely from $E(a)$, $\Omega_K < 0$, and the visible densities. Homogeneous distances therefore determine only the total background residual; a pressureless clustering component and a smooth accelerating component require perturbative observables and a specified closure for perturbations. We give the reconstruction theorem, invariant-closure conditions, closed-distance formulae, compact $S^3$ spectral kernels, an effective-Einstein interpretation for modified gravity, and a falsification protocol using CMB, BAO, supernovae, cosmic chronometers, lensing, growth, clusters, and galaxy dynamics. Compact $S^3$ geometry supplies the finite closed stage; the dark sector is the invariant residual target to be reconstructed covariantly, closed invariantly, and tested observationally.

## I. SCOPE AND CENTRAL STATEMENT

The model class is the compact closed-FLRW spacetime

$$\mathcal{M} = \mathbb{R} \times S^3, \qquad \mathrm{d}s^2 = -N^2(t)c^2\mathrm{d}t^2 + a^2(t)R_0^2\mathrm{d}\Omega_3^2, \tag{1}$$

with $a_0 = 1$ and $R_0$ the present curvature radius. The dark residual is defined by

$$T^{\mathcal{D}}_{\mu\nu} := \frac{c^4}{8\pi G}G_{\mu\nu} - T^{\rm vis}_{\mu\nu}. \tag{2}$$

This is a tensor definition inside general relativity. It is not a gauge choice, a particle model, a proof of a microscopic substance, or a topology-only explanation.

*Nonclaim* I.1 (No topology-only dark sector). The implications

$$S^3 \Longrightarrow \text{dark matter}, \qquad S^3 \Longrightarrow \text{dark energy}$$

are false. Compactness fixes global spatial geometry and infrared spectral structure. It does not by itself imply clustering mass, $\ddot{a} > 0$, $w = -1$, or constant vacuum energy.

*Nonclaim* I.2 (No observationally large curvature assumption). The assumption $\Sigma \simeq S^3$ does not require a curvature radius near the present horizon. The radius $R_0$ may be much larger than current cosmological distance scales, and the reconstruction must be tested against the tight CMB+BAO posterior on $\Omega_K$, which is consistent with flatness in standard analyses [14].

*Nonclaim* I.3 (No tensor-vacuum mechanism). No transverse-traceless vacuum source is used in the present construction. A future tensor source belongs in this framework only after an action or operator, state, renormalization prescription, conserved stress tensor, perturbation equations, and likelihood map are specified.

The strict implication proved here is

$$\begin{gathered} S^3 + \text{closed FLRW} + \text{GR} \\ + \text{visible sector} + \text{specified or reconstructed } E(a) \\ \Longrightarrow T^{\mathcal{D}}_{\mu\nu} \text{ as a covariantly conserved residual.} \end{gathered} \tag{3}$$

The physical interpretation of this residual is then tested by distances, lensing, growth, cluster abundance, and galaxy dynamics.

The novelty is not the algebraic existence of Eq. (2), which follows once the Einstein equation is assumed. The novelty is the closed-geometry bookkeeping: the same compact background, visible-sector closure, curvature convention, distance map, perturbation basis, and falsification ledger are used to define the common target that particle dark matter, dark energy, interacting sectors, or effective modified gravity must reproduce.

*Remark* I.4 (Target residual). The object reconstructed here is not a new substance and not a topology-induced source. It is the covariant target residual that any dark-sector model, interacting sector, or effective modified-gravity model must reproduce once the visible sector, closed geometry, and expansion history are fixed.

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## II. INVARIANT KINEMATICS AND ASSUMPTIONS

The metric signature is $(-,+,+,+)$. The symbol $\epsilon$ denotes energy density and $p$ denotes pressure, with the same dimensions. Let $U^\mu$ be the comoving four-velocity normalized by $U_\mu U^\mu = -c^2$. In the coordinates of Eq. (1), $U^\mu = (1/N, 0, 0, 0)$. The expansion scalar is defined with this dimensional congruence,

$$\Theta := \nabla_\mu U^\mu, \tag{4}$$

so that

$$H = \frac{1}{3}\Theta = \frac{\dot{a}}{Na}, \qquad E(a) = \frac{H(a)}{H_0}. \tag{5}$$

The present critical energy density and curvature parameter are

$$\epsilon_{\mathrm{crit},0} = \frac{3c^2 H_0^2}{8\pi G}, \qquad R_{\mathrm{phys}}(a) = aR_0,$$
$$\Omega_K = -\frac{c^2}{H_0^2 R_0^2} < 0. \tag{6}$$

The sufficient assumptions are:

1. $\Sigma \simeq S^3$ with the round constant-curvature metric in the homogeneous background.
2. General relativity for the total stress tensor: $G_{\mu\nu} = 8\pi G T_{\mu\nu}/c^4$.
3. The visible stress tensor is specified independently and is covariantly conserved unless an explicit interaction term is introduced.
4. The visible sector consists of baryons, photons, and neutrinos with standard thermodynamic and kinetic closure. Baryon density, neutrino masses, and calibration parameters are sampled consistently in the same inference so that no constraint is counted twice.
5. Distances, covariance kernels, lensing, growth, clusters, and galaxy dynamics are evaluated in the same geometry and under the same priors.

No claim of logical minimality is made.

**Proposition II.1** (Invariant expansion function)**.** *In the closed-FLRW class, $E(a) = H(a)/H_0$ is invariant under time reparametrizations once the normalization $a_0 = 1$ is fixed.*

*Proof.* The proper time of a comoving observer is $\mathrm{d}\tau = N(t)\mathrm{d}t$. Hence

$$H = \frac{1}{a}\frac{\mathrm{d}a}{\mathrm{d}\tau}$$

is unchanged when $t$ is replaced by another time coordinate and the lapse is transformed accordingly. Division by the measured present value $H_0$ gives $E(a)$. □

## III. COMPACT $S^3$ GEOMETRY

For the round three-sphere of radius $R$ [1–3],

$$\mathrm{Vol}(S^3_R) = 2\pi^2 R^3, \qquad {}^{(3)}\mathcal{R} = \frac{6}{R^2}, \qquad \mathrm{diam}(S^3_R) = \pi R. \tag{7}$$

A geodesic ball of angular radius $\chi \in [0, \pi]$ has

$$V(\chi) = 2\pi R^3 \left(\chi - \frac{1}{2}\sin 2\chi\right), \qquad A(\chi) = 4\pi R^2 \sin^2\chi. \tag{8}$$

**Lemma III.1** (Compact closure)**.** *$S^3_R$ is compact, orientable, simply connected, boundaryless, and geodesically complete. Its maximum geodesic distance is $\pi R$.*

*Proof.* The round three-sphere is the embedded set $\{x \in \mathbb{R}^4 : |x| = R\}$ with the induced metric. The distance is $d(p,q) = R\arccos[(p\cdot q)/R^2]$. Its maximum occurs at antipodal points. Compact Riemannian manifolds are complete by Hopf–Rinow. □

For the ultrastatic metric, radial null propagation obeys $R\mathrm{d}\chi = c\mathrm{d}t$. The accessible volume fraction after a time interval $T$ is therefore

$$\Gamma(T) = \frac{\theta - \frac{1}{2}\sin 2\theta}{\pi}, \qquad \theta = \min\left(\frac{cT}{R}, \pi\right). \tag{9}$$

For the expanding metric, antipodal causal accessibility between $t_i$ and $t_f$ is

$$\int_{t_i}^{t_f} \frac{c\,N(t)\mathrm{d}t}{a(t)R_0} = \int_{\tau_i}^{\tau_f} \frac{c\,\mathrm{d}\tau}{a(\tau)R_0} \geq \pi. \tag{10}$$

This is causal reach, not a recurrence or topological echo. In unquotiented $S^3$ there are no nontrivial deck transformations.

## IV. COMPACT SPECTRAL COVARIANCE AND PERTURBATIVE BOOKKEEPING

The compact spectral statement is independent of dark-sector microphysics. On a compact boundaryless Riemannian manifold, self-adjoint elliptic Laplace-type operators have compact resolvent, real discrete spectrum, finite-dimensional eigenspaces, and complete $L^2$ eigensections. For the round three-sphere this gives exact infrared discreteness while leaving the ultraviolet tower unbounded.

We label scalar harmonics by the principal $S^3$ level $n$,

$$Y_{n\ell m}, \qquad 0 \leq \ell \leq n, \qquad -\ell \leq m \leq \ell.$$

Here $n$ is the hyperspherical level and $\ell$ is an internal angular label; it must not be confused with the observed sky multipole in a CMB map. The scalar Laplacian satisfies [8]

$$-\nabla^2 Y_{n\ell m} = \frac{n(n+2)}{R^2} Y_{n\ell m},$$

$$g_n^S = (n+1)^2, \qquad n = 0, 1, 2, \ldots. \tag{11}$$

For unit radius the isotropic scalar addition kernel is

$$\Xi_n(\gamma) = \frac{n+1}{2\pi^2} \frac{\sin[(n+1)\gamma]}{\sin\gamma}, \qquad \gamma = d(x, x')/R. \tag{12}$$

With the radius restored, the dimensionful kernel carries the factor $R^{-3}$. An isotropic scalar covariance is therefore

$$C(\gamma) = \sum_{n=0}^{\infty} A_n \, \Xi_n(\gamma). \tag{13}$$

Compactness discretizes the infrared spectrum. It does not impose an ultraviolet cutoff, since $n$ is unbounded.

Linear perturbations on $\mathbb{R} \times S^3$ require an $SO(4)$-adapted scalar–vector–tensor basis rather than a continuum Fourier basis [8–12]. The eigenvalues below use the rough covariant Laplacian $-\nabla^a\nabla_a$ on the round three-sphere. A physical perturbation operator may contain additional gauge, constraint, mass, or curvature-potential terms; such terms must be specified by the model.

A scalar-derived longitudinal vector obeys, for $n \geq 1$,

$$-\nabla^2(\nabla_a Y_{n\ell m}) = \frac{n(n+2)-2}{R^2} \nabla_a Y_{n\ell m}. \tag{14}$$

The independent transverse vector harmonics satisfy

$$\begin{aligned} &\nabla^a Y^{V,\sigma}_{n\ell ma} = 0, \\ &-\nabla^2 Y^{V,\sigma}_{n\ell ma} = \frac{n(n+2)-1}{R^2} Y^{V,\sigma}_{n\ell ma}, \qquad \sigma = 1, 2. \end{aligned} \tag{15}$$

with total transverse-vector degeneracy

$$g_n^{V_T} = 2n(n+2), \qquad n \geq 1. \tag{16}$$

For $n = 1$ these are the six Killing fields of the round three-sphere.

Symmetric rank-two perturbations decompose into scalar-derived, vector-derived, and transverse-traceless parts. The TT harmonics are defined by

$$\nabla^a Y^{T,\sigma}_{n\ell mab} = 0, \qquad g^{ab} Y^{T,\sigma}_{n\ell mab} = 0, \qquad \sigma = 1, 2. \tag{17}$$

With the same rough-Laplacian convention,

$$-\nabla^2 Y^{T,\sigma}_{n\ell mab} = \frac{n(n+2)-2}{R^2} Y^{T,\sigma}_{n\ell mab}, \qquad n \geq 2, \tag{18}$$

and the total TT degeneracy is

$$g_n^{TT} = 2(n-1)(n+3), \qquad n \geq 2. \tag{19}$$

Equivalently, with $j = n + 1$, the rough eigenvalue is $(j^2 - 3)/R^2$ and the TT degeneracy is $2(j^2 - 4)$ for $j \geq 3$.

*Remark* IV.1 (No scalar-spectrum substitution). It is incorrect to assign the scalar eigenvalue $n(n+2)/R^2$ to all perturbation sectors. Vector and tensor harmonics carry curvature shifts, and physical operators may carry further model-dependent terms. A perturbative dark-sector or modified-gravity model must therefore specify its actual operator before claiming predictions for low-$n$ modes.

The compact replacement rule for perturbative bookkeeping is

$$\int \frac{\mathrm{d}^3k}{(2\pi)^3} \mathcal{F}(k) \quad \longrightarrow \quad \frac{1}{\mathrm{Vol}(S^3_R)} \sum_{n\in\mathcal{N}} \sum_{\alpha=1}^{g_n} \mathcal{F}_{n\alpha}. \tag{20}$$

The degeneracy index $\alpha$ belongs to the relevant scalar, vector, or tensor sector, and $\mathcal{N}$ excludes homogeneous and gauge subspaces according to the perturbation variable. This is mode bookkeeping, not a claim that compactness removes every infrared subtlety. Gauge zero modes, state-dependent long-wavelength effects, and de Sitter-like infrared questions remain operator- and state-dependent.

For CMB temperature anisotropies, a scalar harmonic restricted to the last-scattering sphere has the separated form

$$Y_{n\ell m}(\chi_*, \theta, \phi) = H_{n\ell}(\chi_*) Y_{\ell m}(\theta, \phi), \tag{21}$$

so a closed-universe line-of-sight calculation replaces the flat Fourier integral by a hyperspherical sum,

$$C_\ell^{TT} = \sum_{n\in\mathcal{N}_S,\, n\geq\ell} \mathcal{W}_{n\ell} \, \mathcal{P}_{\mathcal{R}}(n) \left|\Delta^T_{\ell n}\right|^2. \tag{22}$$

This schematic expression is written up to the normalization convention used for the hyperspherical harmonics, degeneracy weights, primordial spectrum, and transfer functions. Here $\ell$ is the observed sky multipole. The transfer functions $\Delta^T_{\ell n}$ must be computed in the same closed geometry and with the same visible-sector and residual-sector parameters used in the background likelihood; public Boltzmann codes such as CAMB or CLASS implement this when $\Omega_K \neq 0$ is specified [37, 38]. Equation (22) is deliberately conservative: compactness supplies a discrete infrared spectrum, but the sign and size of any low-$\ell$ change depend on the primordial spectrum, transfer functions, ISW terms, reionization, foreground treatment, estimator, mask, and likelihood [13, 15].

## V. CLOSED-FLRW BACKGROUND RECONSTRUCTION

With curvature kept on the geometric side, the Friedmann equation is [1, 4, 5, 7]

$$H^2 = \frac{8\pi G}{3c^2} \epsilon_{\mathrm{tot}} - \frac{c^2}{a^2 R_0^2}. \tag{23}$$

Equivalently,

$$\frac{\epsilon_{\mathrm{tot}}(a)}{\epsilon_{\mathrm{crit},0}} = E^2(a) - \Omega_K a^{-2}. \tag{24}$$

Conservation gives

$$\dot{\epsilon}_{\mathrm{tot}} + 3H(\epsilon_{\mathrm{tot}} + p_{\mathrm{tot}}) = 0, \tag{25}$$

and hence

$$\frac{p_{\rm tot}(a)}{\epsilon_{\rm crit,0}} = -\frac{1}{3}\left[\frac{\mathrm{d}E^2}{\mathrm{d}\ln a} + 3E^2(a) - \Omega_K a^{-2}\right]. \tag{26}$$

The deceleration scalar is

$$q(a) := -\frac{\ddot{a}\,a}{\dot{a}^2} = -1 - \frac{1}{2}\frac{\mathrm{d}\ln E^2}{\mathrm{d}\ln a}. \tag{27}$$

The acceleration equation is

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}\left(\epsilon_{\rm tot} + 3p_{\rm tot}\right). \tag{28}$$

Thus accelerated expansion is the scalar inequality

$$\epsilon_{\rm tot} + 3p_{\rm tot} < 0. \tag{29}$$

Define

$$\mathfrak{A}(a) := -\frac{1}{2}\frac{\epsilon_{\rm tot} + 3p_{\rm tot}}{\epsilon_{\rm crit,0}} = -q(a)E^2(a). \tag{30}$$

Acceleration is equivalent to $\mathfrak{A}(a) > 0$.

## VI. VISIBLE-SECTOR CLOSURE

The visible tensor is

$$T^{\rm vis}_{\mu\nu} = T^{b}_{\mu\nu} + T^{\gamma}_{\mu\nu} + T^{\nu}_{\mu\nu}. \tag{31}$$

On the homogeneous background,

$$\begin{aligned}\frac{\epsilon_{\rm vis}(a)}{\epsilon_{\rm crit,0}} &= \Omega_{\rm b}a^{-3} + \Omega_\gamma a^{-4} + \Omega_\nu(a),\\ \frac{p_{\rm vis}(a)}{\epsilon_{\rm crit,0}} &= \frac{1}{3}\Omega_\gamma a^{-4} + P_\nu(a).\end{aligned} \tag{32}$$

The thermodynamic photon and neutrino densities are evaluated in the standard local kinetic description used by cosmological Boltzmann solvers. Exact finite-volume mode discreteness of quantum fields on a compact $S^3$ would be a separate state- and operator-dependent correction to the visible stress tensor; it is not part of the dark residual unless it is explicitly added to $T^{\rm vis}_{\mu\nu}$.

The baryon term is the rest-energy density of conserved nonrelativistic baryon number,

$$\epsilon_b(a) = \epsilon_{\rm crit,0}\,\Omega_{\rm b}a^{-3}. \tag{33}$$

The photon term is fixed by the observed CMB temperature [29],

$$\begin{aligned}\epsilon_\gamma(a) &= \frac{\pi^2}{15}\frac{(k_{\rm B}T_{\gamma0})^4}{\hbar^3c^3}\,a^{-4} = \epsilon_{\rm crit,0}\,\Omega_\gamma a^{-4},\\ p_\gamma(a) &= \frac{1}{3}\epsilon_\gamma(a).\end{aligned} \tag{34}$$

### A. Massive-neutrino functions

The neutrino temperature and phase-space form follow from standard weak decoupling, entropy conservation, and collisionless propagation [5, 6, 33, 34]. The absolute neutrino mass scale is not fixed by thermodynamics alone; it is an input or nuisance parameter constrained by laboratory and cosmological priors.

At temperatures below the $W$-boson mass, the interaction rate scales as

$$\Gamma_\nu(T) \simeq C_\nu G_F^2 T^5, \qquad C_\nu = \mathcal{O}(1), \tag{35}$$

while during radiation domination

$$H(T) = 1.66\, g_*^{1/2}\frac{T^2}{M_{\rm Pl}}. \tag{36}$$

Freeze-out occurs when $\Gamma_\nu \simeq H$, giving

$$T_{\rm dec} \simeq \left(\frac{1.66\, g_*^{1/2}}{C_\nu G_F^2 M_{\rm Pl}}\right)^{1/3} = \mathcal{O}(1)\ \text{MeV} \tag{37}$$

for $g_* = 10.75$ near the MeV epoch. Neutrinos therefore decouple while ultrarelativistic.

After decoupling, entropy transfer from $e^\pm$ annihilation heats photons relative to neutrinos. In the instantaneous-decoupling approximation,

$$g^{\rm em}_{*s,\rm before} = 2 + \frac{7}{8}\times 4 = \frac{11}{2}, \qquad g^{\rm em}_{*s,\rm after} = 2, \tag{38}$$

and

$$T_{\nu0} = \left(\frac{4}{11}\right)^{1/3} T_{\gamma0} \simeq 1.95\ \text{K}. \tag{39}$$

Non-instantaneous decoupling, finite-temperature QED corrections, and flavor oscillations are commonly summarized by the Standard-Model value $N_{\rm eff} \simeq 3.044$ [30–32]. The analytic formulae below use the instantaneous-decoupling Fermi–Dirac form. In numerical likelihoods this ideal distribution should be replaced by Boltzmann-solver neutrino transfer functions or tabulated distorted spectra consistent with the adopted $N_{\rm eff}$. The tensor residual formulae below are unchanged.

In Eqs. (41)–(42) we use natural units $\hbar = c = k_{\rm B} = 1$. For each mass eigenstate $m_i$, let

$$f_0(q) = \frac{1}{\exp(q/T_{\nu0}) + 1} \tag{40}$$

be the frozen occupation number as a function of comoving momentum $q = ap$, with vanishing chemical potential. With degeneracy $g_i = 2$ for the neutrino plus antineutrino helicity states of one mass eigenstate,

$$\epsilon_{\nu i}(a) = \frac{g_i}{2\pi^2 a^4}\int_0^\infty \mathrm{d}q\, q^2\sqrt{q^2 + a^2m_i^2}\, f_0(q), \tag{41}$$

$$p_{\nu i}(a) = \frac{g_i}{6\pi^2 a^4}\int_0^\infty \mathrm{d}q\,\frac{q^4}{\sqrt{q^2+a^2m_i^2}}\,f_0(q). \tag{42}$$

The total neutrino contribution is

$$\epsilon_\nu(a) = \sum_{i=1}^{3}\epsilon_{\nu i}(a), \qquad p_\nu(a) = \sum_{i=1}^{3} p_{\nu i}(a), \tag{43}$$

and

$$\Omega_\nu(a) = \frac{\epsilon_\nu(a)}{\epsilon_{\mathrm{crit},0}}, \qquad P_\nu(a) = \frac{p_\nu(a)}{\epsilon_{\mathrm{crit},0}}. \tag{44}$$

The limiting checks are

$$am_i \ll q: \quad \epsilon_{\nu i} \propto a^{-4}, \qquad p_{\nu i} = \frac{1}{3}\epsilon_{\nu i}, \tag{45}$$

$$am_i \gg q: \quad \epsilon_{\nu i} \simeq m_i n_{\nu i} \propto a^{-3}, \qquad p_{\nu i} \ll \epsilon_{\nu i}. \tag{46}$$

At late times, for nonrelativistic neutrinos,

$$\Omega_{\nu 0}h^2 \simeq \frac{\sum_i m_i}{93.14\ \mathrm{eV}}. \tag{47}$$

Since the post-decoupling distribution is collisionless,

$$\dot\epsilon_\nu + 3H(\epsilon_\nu + p_\nu) = 0. \tag{48}$$

## VII. GR DARK RESIDUAL

**Theorem VII.1** (GR residual theorem)**.** *Within general relativity, assume the total stress tensor obeys the Einstein equation and the visible stress tensor is covariantly conserved, $\nabla_\mu T^{\mathrm{vis}\,\mu\nu} = 0$. Then the tensor $T^{\mathcal{D}}_{\mu\nu}$ defined by Eq.* (2) *is symmetric and covariantly conserved. On the closed-FLRW background it has the perfect-fluid form*

$$T^{\mathcal{D}}_{\hat\mu\hat\nu} = \mathrm{diag}(\epsilon_{\mathcal{D}}, p_{\mathcal{D}}, p_{\mathcal{D}}, p_{\mathcal{D}})$$

*in the comoving orthonormal frame, with*

$$\frac{\epsilon_{\mathcal{D}}(a)}{\epsilon_{\mathrm{crit},0}} = E^2(a) - \Omega_K a^{-2} - \Omega_{\mathrm{b}} a^{-3} - \Omega_\gamma a^{-4} - \Omega_\nu(a), \tag{49}$$

$$\frac{p_{\mathcal{D}}(a)}{\epsilon_{\mathrm{crit},0}} = -\frac{1}{3}\left[\frac{\mathrm{d}E^2}{\mathrm{d}\ln a} + 3E^2(a) - \Omega_K a^{-2}\right] - \frac{1}{3}\Omega_\gamma a^{-4} - P_\nu(a). \tag{50}$$

*Proof.* Both $G_{\mu\nu}$ and $T^{\mathrm{vis}}_{\mu\nu}$ are symmetric rank-two tensors; their linear combination is therefore a symmetric rank-two tensor. The twice-contracted Bianchi identity gives $\nabla_\mu G^{\mu\nu} = 0$, hence

$$\nabla_\mu T^{\mathcal{D}\,\mu\nu} = \frac{c^4}{8\pi G}\nabla_\mu G^{\mu\nu} - \nabla_\mu T^{\mathrm{vis}\,\mu\nu} = 0.$$

Homogeneity and isotropy force the background part of any covariantly defined symmetric tensor to be a linear combination of $U_\mu U_\nu$ and the spatial projector $h_{\mu\nu} := g_{\mu\nu} + c^{-2}U_\mu U_\nu$, by Schur's lemma applied to the isotropy group SO(3); it is therefore a perfect fluid. The 00 Einstein equation gives Eq. (24). Subtracting Eq. (32) gives Eq. (49). For pressure, Eq. (25) gives $p_{\mathrm{tot}} = -\epsilon_{\mathrm{tot}} - \frac{1}{3}\mathrm{d}\epsilon_{\mathrm{tot}}/\mathrm{d}\ln a$, which is Eq. (26). Subtracting the visible pressure gives Eq. (50). □

*Remark* VII.2 (Curvature is not relabeled as a dark fluid)*.* The curvature term is kept as geometry through $\Omega_K$ and is not counted as dark matter or dark energy. A curvature-only explanation would require the residual reconstructed by Eqs. (49) and (50) to vanish or become observationally unnecessary in a global likelihood analysis.

The residual equation of state, where defined, is

$$w_{\mathcal{D}}(a) = \frac{p_{\mathcal{D}}(a)}{\epsilon_{\mathcal{D}}(a)} = \frac{P_{\mathcal{D}}(a)}{\Omega_{\mathcal{D}}(a)}. \tag{51}$$

*Remark* VII.3 (Residual is not a microphysical explanation)*.* Equation (2) reconstructs what is missing from the specified visible sector within GR. It does not distinguish particle dark matter, vacuum energy, modified gravity written as an effective tensor, or an interacting dark sector. That distinction requires perturbations and model comparison.

## VIII. CLUSTERING RESIDUAL AND SMOOTH RESIDUAL

The homogeneous background cannot uniquely split the residual into dark matter and dark energy.

**Theorem VIII.1** (Background degeneracy)**.** *On an FLRW background, any residual density $\epsilon_{\mathcal{D}}(a)$ can be decomposed as $\epsilon_{\mathcal{D}}(a) = \epsilon_c(a) + \epsilon_X(a)$ in infinitely many ways unless additional information fixes clustering, equation of state, sound speed, anisotropic stress, or interactions. Homogeneous distances alone do not identify dark matter.*

*Proof.* Choose any differentiable function $\epsilon_c(a)$ and define $\epsilon_X = \epsilon_{\mathcal{D}} - \epsilon_c$. The same total homogeneous residual is obtained. Distances see only the total background expansion. Perturbative observables are required to separate components. □

The phenomenological split used for comparison with standard cosmological likelihoods is

$$\Omega_c(a) = \Omega_{c0}a^{-3}, \qquad p_c = 0, \tag{52}$$

$$\Omega_X(a) = E^2(a) - \Omega_K a^{-2} - \Omega_{\mathrm{b}}a^{-3} - \Omega_\gamma a^{-4} - \Omega_\nu(a) - \Omega_{c0}a^{-3}, \tag{53}$$

$$P_X(a) = -\frac{1}{3}\left[\frac{\mathrm{d}E^2}{\mathrm{d}\ln a} + 3E^2(a) - \Omega_K a^{-2}\right] - \frac{1}{3}\Omega_\gamma a^{-4} - P_\nu(a), \tag{54}$$

$$w_X(a) = \frac{P_X(a)}{\Omega_X(a)}. \tag{55}$$

Here $\Omega_{c0}$ is not determined by the background alone. Within GR and a specified perturbation closure it is constrained by CMB peak structure, CMB lensing, large-scale-structure growth, weak lensing, cluster abundance, and galaxy-scale dynamical mass diagnostics [14, 22–25].

For the smooth residual define

$$\mathfrak{D}_X(a) := \Omega_X(a) + 3P_X(a). \tag{56}$$

The smooth component carries a negative contribution to the total $\epsilon_{\rm tot} + 3p_{\rm tot}$ when $\mathfrak{D}_X(a) < 0$; the sign of the total acceleration is fixed only after matter and radiation are included in Eq. (28).

A cosmological constant is the special case

$$\begin{aligned} T^X_{\mu\nu} &= -\epsilon_\Lambda g_{\mu\nu}, \quad \Omega_X(a) = \Omega_\Lambda, \\ P_X &= -\Omega_\Lambda, \qquad\qquad w_X = -1, \quad \mathfrak{D}_X = -2\Omega_\Lambda. \end{aligned} \tag{57}$$

A dynamical dark-energy reconstruction requires a statistically significant failure of Eq. (57) after curvature, neutrino masses, calibration, and selection effects are marginalized [18, 36]. Current DESI DR2 combinations motivate such tests, but they do not by themselves establish a physical departure from ΛCDM [17, 18].

## IX. OBSERVABLE DISTANCES AND PERTURBATIVE INVARIANTS

For $z = a^{-1} - 1$,

$$D_C(z) = c\int_0^z \frac{\mathrm{d}z'}{H(z')}, \qquad \chi(z) = \frac{D_C(z)}{R_0}. \tag{58}$$

The closed transverse comoving distance and the standard angular and luminosity distances are

$$\begin{aligned} D_M(z) &= R_0 \sin\chi(z), \\ D_A(z) &= \frac{D_M(z)}{1+z}, \qquad D_L(z) = (1+z)D_M(z). \end{aligned} \tag{59}$$

BAO, supernovae, and CMB distances constrain $E(a)$ and $\Omega_K$ through these scalar distances [14, 16, 17, 19, 20]. Common BAO combinations are

$$\begin{aligned} D_H(z) &= \frac{c}{H(z)}, \qquad D_V(z) = \left[zD_M^2(z)D_H(z)\right]^{1/3}, \\ F_{\rm AP}(z) &= \frac{D_M(z)H(z)}{c}. \end{aligned} \tag{60}$$

Etherington reciprocity gives

$$D_L(z) = (1+z)^2 D_A(z) \tag{61}$$

when photons travel on null geodesics and photon number is conserved [28].

The linear growth observables are

$$f(a) = \frac{\mathrm{d}\ln D_+(a)}{\mathrm{d}\ln a}, \qquad f\sigma_8(z), \tag{62}$$

as measured by redshift-space distortions. Weak lensing and CMB lensing measure projected Weyl-potential combinations along null geodesics [22, 23]. Merging clusters supply a complementary diagnostic: offsets between lensing mass and the dominant X-ray plasma are evidence for a gravitating component displaced from the main baryonic gas, not by themselves a proof of a particular particle model [24].

## X. COMPLETE INVARIANT CLOSURE OF THE DARK-SECTOR TARGET

The phrase "closing the dark sector" has two logically different meanings. The first is an exact reconstruction of the stress tensor required by the metric and visible sector. The second is a microphysical explanation of that tensor in terms of particles, fields, vacuum energy, interactions, or modified gravity. Only the first is model-independent. The second requires extra dynamical structure and cannot be obtained from topology or background geometry alone.

**Definition X.1** (Invariant residual closure)**.** Fix the closed-FLRW background, the visible stress tensor, and a perturbative order. The dark residual is invariantly closed at that order if, for every scalar, vector, and tensor harmonic coefficient retained in the calculation,

$$T^{\mathcal{D}}_{(r)\mu\nu} = \frac{c^4}{8\pi G}G_{(r)\mu\nu} - T^{\rm vis}_{(r)\mu\nu}, \qquad r = 0, 1, \ldots, N, \tag{63}$$

is specified in a gauge-invariant representation, and the same variables are used in the distance, lensing, growth, CMB, cluster, and galaxy likelihoods. Here $r = 0$ denotes the homogeneous background and $r \geq 1$ denotes perturbative orders.

At linear scalar order one may choose the Bardeen potentials, matter velocity potentials, density contrasts, and anisotropic stresses. In the closed model each of these is expanded in hyperspherical harmonics rather than Fourier modes. Let $\mathcal{T}^{\mathcal{D}}_{n\ell m}$ denote the scalar coefficient tuple $\{\delta\epsilon^{\mathcal{D}}_{n\ell m}, \delta p^{\mathcal{D}}_{n\ell m}, (\epsilon_{\mathcal{D}} + p_{\mathcal{D}})V^{\mathcal{D}}_{n\ell m}, \Pi^{\mathcal{D}}_{n\ell m}\}$, and let $\mathcal{G}_{n\ell m}$ and $\mathcal{T}^{\rm vis}_{n\ell m}$ denote the corresponding Einstein and visible-sector tuples. Then

$$\mathcal{T}^{\mathcal{D}}_{n\ell m} = \frac{c^4}{8\pi G}\mathcal{G}_{n\ell m} - \mathcal{T}^{\rm vis}_{n\ell m}. \tag{64}$$

Equation (64) is not a new field equation. It is the harmonic version of the tensor identity Eq. (2). Vector and tensor coefficients are obtained in the corresponding transverse-vector and transverse-traceless bases of Sec. IV.

**Theorem X.2** (No purely geometric closure of the dark sector)**.** *No collection of invariants constructed only from the topology of $S^3$, the round spatial curvature radius, the compact Laplace spectra, the homogeneous distance map, or their algebraic combinations can determine a unique physical split of $T^{\mathcal{D}}_{\mu\nu}$ into dark matter and dark energy.*

*Proof.* Those invariants determine the stage on which the metric and perturbations are expanded. They do not determine an equation of state, sound speed, anisotropic stress, interaction current, rest frame, or clustering law for an unknown component. On the homogeneous background the degeneracy theorem already gives infinitely many decompositions $\epsilon_{\mathcal{D}} = \epsilon_c + \epsilon_X$. At perturbative order the same residual metric response can be represented by different combinations of density contrast, pressure perturbation, anisotropic stress, and modified-gravity response functions unless a dynamical closure is imposed. Therefore geometry alone cannot supply the missing microphysics. □

**Theorem X.3** (Complete covariant target closure)**.** *Within GR, given the metric to the chosen perturbative order and a separately specified visible sector, Eq.* (63) *uniquely closes the dark residual as a tensor. This is the maximal model-independent closure: every particle dark-sector model, interacting-sector model, vacuum model, or effective modified-gravity model must reproduce these residual coefficients in the same closed geometry.*

*Proof.* The Einstein tensor is determined by the metric order by order. The visible stress tensor is specified independently. Their difference is a unique tensor coefficient in each scalar, vector, and tensor harmonic sector. Gauge-invariant variables give the same tensor coefficients in any coordinate representation. Any model producing the same metric and visible sector must therefore produce the same residual tensor, or the same effective tensor if it is written in Einstein form. □

A compact way to display the complete invariant target is

$$\mathcal{C}_{\rm inv} = \left[\Omega_K,\ E(a),\ T^{\rm vis}_{\mu\nu};\ T^{\mathcal{D}}_{\mu\nu};\ \{\delta T^{\mathcal{D}}_{n\alpha}\}_{S,V,T};\ \mathcal{R}_{\rm dyn}\right]. \tag{65}$$

The first entries fix the closed background and visible sector. The middle entries are the exact background and perturbative residuals. The final entry $\mathcal{R}_{\rm dyn}$ is not determined by geometry; it denotes the dynamical response required to interpret the residual as dust, a smooth component, vacuum energy, an interacting sector, or modified gravity. In a likelihood analysis the corresponding posterior has the schematic invariant factorization

$$\mathcal{P}(\mathcal{C}_{\rm inv} \mid \mathcal{D}) \propto \pi(\mathcal{C}_{\rm inv})\, \mathcal{L}_{\rm CMB}\mathcal{L}_{\rm BAO}\mathcal{L}_{\rm SN} \times \mathcal{L}_{H(z)}\mathcal{L}_{\rm lens}\mathcal{L}_{\rm growth}\mathcal{L}_{\rm cl}\mathcal{L}_{\rm gal}. \tag{66}$$

where all likelihoods use the same curvature, visible-sector priors, and closed-harmonic perturbation convention. Observational closure means that the same posterior samples pass all of these tests without moving curvature, neutrino mass, calibration, or selection effects between inconsistent priors.

The invariant closure filters that have physical content are the following.

1. *Pressureless clustering filter.* A component $c$ qualifies as cold clustering matter only if
$$T^c_{\mu\nu} = \frac{\epsilon_c}{c^2} u^c_\mu u^c_\nu, \qquad u^c_\mu u^\mu_c = -c^2,$$
$$p_c = 0, \qquad \Pi^c_{\mu\nu} = 0,$$
$$\nabla_\mu T^{c\,\mu\nu} = Q^\nu_c. \tag{67}$$
with a stated interaction current $Q^\nu_c$ and with $Q^\nu_c = 0$ in the separately conserved case. In the non-interacting limit the flow is geodesic where $\epsilon_c \neq 0$. This filter is tested by CMB peak structure, lensing, growth, cluster abundance, and galaxy dynamics.

2. *Smooth accelerating filter.* The residual left after the clustering component is $T^X_{\mu\nu} = T^{\mathcal{D}}_{\mu\nu} - T^c_{\mu\nu}$. It contributes negatively to the Raychaudhuri source when
$$\mathfrak{D}_X(a) = \Omega_X(a) + 3P_X(a) < 0. \tag{68}$$
Calling it smooth is an observational and dynamical statement: the perturbation closure must specify its rest-frame sound speed, anisotropic stress, and any interaction current, or show that its clustering is negligible in the relevant likelihood.

3. *Cosmological-constant filter.* The smooth component is exactly a cosmological constant if and only if
$$T^X_{\mu\nu} = -\epsilon_\Lambda g_{\mu\nu}, \qquad \nabla_\alpha \epsilon_\Lambda = 0, \tag{69}$$
so that all gauge-invariant perturbations of this component vanish. In the background language this is the joint condition $\Omega_X = \Omega_\Lambda$, $P_X = -\Omega_\Lambda$, and $w_X = -1$ at all $a$.

4. *Modified-gravity filter.* If the residual is moved to the right-hand side of an effective Einstein equation, the background conditions above are necessary but insufficient. The model must also provide invariant metric-response functions, or their closed-harmonic equivalents, that reproduce both the dynamical potential and the Weyl potential probed by growth and lensing. In phenomenological notation these are often represented by functions such as $\mu(a, n)$, $\Sigma(a, n)$, and a gravitational-slip variable.

Thus the exhaustive invariant answer is narrow but precise. Topology and compact spectra close the geometry. The Einstein residual closes the required stress tensor. A physical dark-sector explanation is closed only after one of the dynamical filters above supplies the missing perturbative response and passes the joint likelihood.

## XI. DIAGNOSTIC LEDGER

The following quantities form a compact diagnostic ledger. They are not independent axioms; they are invariant or operational quantities used to reconstruct and test the residual.

1. $\Sigma \simeq S^3$: compact closed spatial topology.
2. $\partial S^3 = \varnothing$: no spatial boundary source in the unquotiented model.
3. $\pi_1(S^3) = 0$: no holonomy images or deck transformations.
4. $V_3 = 2\pi^2 R^3$: finite spatial volume at fixed scale factor.
5. $\mathrm{diam}(S^3_R) = \pi R$: finite diameter.
6. ${}^{(3)}\mathcal{R} = 6/R^2$: positive spatial curvature scalar.
7. $\Omega_K = -c^2/(H_0^2 R_0^2)$: closed curvature parameter.
8. $\Theta = \nabla_\mu U^\mu$: invariant expansion scalar.
9. $E(a) = H/H_0$: dimensionless expansion history.
10. $q(a)$ and $\mathfrak{A}(a)$: acceleration diagnostics.
11. $\epsilon_{\mathrm{tot}}/\epsilon_{\mathrm{crit},0} = E^2 - \Omega_K a^{-2}$: total density reconstruction.
12. $p_{\mathrm{tot}}/\epsilon_{\mathrm{crit},0}$: total pressure reconstruction.
13. $T^{\mathcal{D}}_{\mu\nu}$: covariant dark residual.
14. $\epsilon_{\mathcal{D}}(a), p_{\mathcal{D}}(a)$: homogeneous residual scalars.
15. $\nabla_\mu T^{\mathcal{D}\,\mu\nu} = 0$: Bianchi consistency under visible conservation.
16. $\Omega_b h^2$: baryon density constrained by CMB and BBN.
17. $T_{\gamma 0}$: photon temperature fixing $\Omega_\gamma$.
18. $T_{\nu 0}$ and $f_0(q)$: neutrino thermal closure.
19. $\sum_i m_i$: neutrino mass-sum control parameter.
20. $\Omega_c a^{-3}, p_c = 0$: pressureless clustering ansatz.
21. CMB peak structure: matter loading and equality diagnostics.
22. $C_L^{\phi\phi}$: CMB lensing mass map.
23. Weak shear $\xi_\pm$: projected late-time mass correlations.
24. $f\sigma_8(z)$: growth-rate observable.
25. Cluster abundance: mass-function test.
26. Galaxy dynamics: small-scale mass-discrepancy diagnostic.
27. $\Omega_X(a), P_X(a)$: smooth residual after clustering matter.
28. $w_X(a)$: smooth equation-of-state reconstruction.
29. $\mathfrak{D}_X = \Omega_X + 3P_X$: smooth acceleration factor.
30. $D_M, D_A, D_L, D_H, D_V, F_{\mathrm{AP}}$: closed-distance and BAO observables.
31. $\theta_* = r_s(z_*)/D_M(z_*)$: CMB acoustic-angle constraint.
32. $\lambda_n = n(n+2)/R^2$: compact scalar spectrum.
33. $g_n^S = (n+1)^2$: scalar mode count.
34. $\Xi_n(\gamma)$: addition-kernel invariant.
35. $C(\gamma) = \sum_n A_n \Xi_n$: compact scalar covariance.
36. $g_n^{V_T} = 2n(n+2)$: transverse-vector mode count.
37. $g_n^{TT} = 2(n-1)(n+3)$: transverse-traceless tensor mode count.
38. Eq. (20): compact perturbation replacement rule.
39. Eq. (22): closed-geometry CMB transfer-sum interface.
40. Eq. (65): minimal complete invariant target.
41. Eq. (63): order-by-order invariant residual closure.
42. Eq. (66): joint posterior closure criterion.
43. Eqs. (67)–(69): clustering, acceleration, and cosmological-constant filters.
44. Posterior predictive residuals: falsification beyond best fit.
45. Flat metric limit $R_0 \to \infty$: local reduction to standard FLRW distances.

## XII. DATA-FACING RECONSTRUCTION PROTOCOL

*Invariant reconstruction protocol.*

1. Choose the geometry, field equation, priors, and $E(a)$ basis.
2. Reconstruct $E(a)$ and $\Omega_K$ from CMB, BAO, supernovae, and chronometers [21].
3. Fix $\Omega_b h^2$, $\Omega_\gamma$, and the neutrino functions $\Omega_\nu(a), P_\nu(a)$ conditional on the chosen mass spectrum, BBN/CMB constraints, laboratory priors, and jointly sampled cosmological nuisance priors [26, 27].
4. Compute $\epsilon_{\mathcal{D}}(a)$ and $p_{\mathcal{D}}(a)$ from Eqs. (49) and (50).
5. Apply the invariant closure filters of Sec. X: residual-only closure, dust-like clustering closure, smooth-acceleration closure, cosmological-constant closure, or effective modified-gravity closure.

6. Infer $\Omega_{c0}$ from perturbative observables: CMB peaks, CMB lensing, weak lensing, $f\sigma_8$, clusters, and galaxy dynamics.

7. Compute $\Omega_X(a)$, $P_X(a)$, $w_X(a)$, and $\mathfrak{D}_X(a)$ from Eqs. (53)–(56).

8. Compare $\Omega_X = \text{const}$, $w_X = -1$, evolving $w(z)$, interacting dark sectors, modified-gravity effective tensors, and curvature-only explanations under the same likelihoods.

9. Build compact $S^3$ scalar, vector, and tensor mode expansions using Eq. (20); compare low-multipole CMB covariance with the same curvature posterior.

10. Report posterior predictive checks, evidence or information criteria, and the local flat-metric limit $R_0 \to \infty$.

Finite noisy data reconstruct a posterior over $E(a)$, not a unique curve. The uniqueness statements are conditional on a specified geometry, visible sector, and exact or posterior-sampled expansion history.

## XIII. FORMAL CONSEQUENCES

**Theorem XIII.1** (Invariant reconstruction theorem)**.** *Given a closed-FLRW geometry, the GR field equation, a covariantly conserved visible stress tensor, and an expansion history $E(a)$ with curvature $\Omega_K$, the residual $T^{\mathcal{D}}_{\mu\nu}$ is uniquely determined as a tensor by Eq.* (2)*. Its background energy density and pressure are the scalars in Eqs.* (49) *and* (50)*.*

*Proof.* The metric determines the Einstein tensor. The visible tensor is specified independently. Their difference is a unique tensor. FLRW symmetry reduces its background part to two scalar functions, computed above. □

**Theorem XIII.2** (Perturbative separation theorem)**.** *Within GR and a specified perturbation closure, a pressureless clustering component and a smooth accelerating component cannot be distinguished by homogeneous distances alone. They can be separated only by perturbative observables such as CMB peak structure, lensing, growth, cluster abundance, and galaxy-scale dynamical mass diagnostics.*

*Proof.* Homogeneous distances constrain $E(a)$ and therefore only the total homogeneous density and pressure. Perturbations depend on clustering, sound speed, anisotropic stress, and gravitational potentials. These quantities affect lensing and growth differently from a smooth cosmological-constant-like component. Hence the split is observationally meaningful only after perturbative data and a perturbation model are specified. □

## XIV. EFFECTIVE INTERPRETATION IN MODIFIED GRAVITY

The residual construction is not tied to a particle interpretation. It also gives the target effective tensor for any metric modified-gravity model that can be written in Einstein form with separately conserved visible matter.

**Proposition XIV.1** (Effective Einstein form)**.** *Suppose a diffeomorphism-covariant metric theory has field equations that can be rearranged as*

$$G_{\mu\nu} = \frac{8\pi G}{c^4}\left(T^{\text{vis}}_{\mu\nu} + T^{\text{eff}}_{\mu\nu}\right), \tag{70}$$

*with $\nabla_\mu T^{\text{vis}\,\mu\nu} = 0$. Then the effective stress tensor required by the modified theory is exactly the covariant residual,*

$$T^{\text{eff}}_{\mu\nu} = T^{\mathcal{D}}_{\mu\nu}. \tag{71}$$

*Consequently its homogeneous energy density and pressure must equal Eqs.* (49) *and* (50)*.*

*Proof.* Equation (70) gives

$$T^{\text{eff}}_{\mu\nu} = \frac{c^4}{8\pi G}G_{\mu\nu} - T^{\text{vis}}_{\mu\nu},$$

which is precisely Eq. (2). Conservation follows from the Bianchi identity and visible-sector conservation exactly as in the GR residual theorem. □

The proposition is an identity, not a new dynamical model. For example, an $f(R)$ model can be moved into effective Einstein form by placing the extra curvature terms on the right-hand side [35]. The homogeneous target is then fixed:

$$\epsilon_{\text{eff}}(a) = \epsilon_{\mathcal{D}}(a), \qquad p_{\text{eff}}(a) = p_{\mathcal{D}}(a).$$

A model that cannot reproduce these functions for the reconstructed $E(a)$ and $\Omega_K$ fails at the homogeneous level.

The equivalence is conditional. If visible matter is nonminimally coupled so that $\nabla_\mu T^{\text{vis}\,\mu\nu} \neq 0$, or if photons do not follow the metric null geodesics used in Eq. (59), then the present reconstruction theorem must be modified. The interaction current, altered distance law, and perturbation equations must then be stated explicitly.

At the perturbative level, matching the background residual is insufficient. A viable modified-gravity explanation must also reproduce the gravitational potentials probed by CMB lensing, weak lensing, redshift-space distortions, clusters, and galaxy dynamics. In phenomenological language, the model must provide the analogues of the effective Newton response and lensing response, often denoted $\mu(a,k)$ and $\Sigma(a,k)$, together with any anisotropic stress, sound speed, or extra-field propagation law. These perturbative functions are not determined by Eqs. (49)–(50); they are additional model predictions.

Compactness adds a global consistency requirement. Perturbations on the unquotiented round $S^3$ use the discrete eigenvalues and kernels above, not a continuous $\mathbb{R}^3$ Fourier basis. Nonlocal kernels, infrared prescriptions, and vacuum choices in a modified-gravity model must therefore be defined on the compact spectrum and must reduce continuously to the local flat-space expressions as $R_0 \to \infty$.

## XV. FALSIFICATION CRITERIA

The compact reconstruction program fails, or becomes less useful than a simpler parametrization, if any of the following occur:

1. Closed-FLRW distances are disfavored relative to flat or curved ΛCDM after equal-prior comparison.
2. The reconstructed $T^{\mathcal{D}}_{\mu\nu}$ violates conservation without an explicit interaction term.
3. Perturbative data do not admit a stable pressureless clustering component within the stated model class.
4. The inferred smooth component is inconsistent across CMB, BAO, supernovae, lensing, and growth.
5. Compact $S^3$ covariance worsens posterior predictive tests.
6. Apparent dark-energy evolution disappears after curvature, neutrino mass, calibration, and selection-function marginalization.
7. Visible-sector closure is inconsistent with BBN, CMB thermodynamics, or neutrino priors.
8. A modified-gravity interpretation cannot supply an effective tensor and perturbation sector matching the reconstructed residual, lensing, and growth observables.
9. The local large-radius limit does not reduce continuously to the standard FLRW expressions.

## XVI. CONCLUSION

The strict compact-geometric result is

$$\mathbb{R}\times S^3 \Longrightarrow \begin{array}{c}\text{finite closed geometry}\\ \text{and compact spectral covariance.}\end{array} \tag{72}$$

The strict GR residual result is

$$G_{\mu\nu}-\frac{8\pi G}{c^4}T^{\mathrm{vis}}_{\mu\nu} \Longrightarrow \begin{array}{c}T^{\mathcal{D}}_{\mu\nu}\text{ as a covariantly}\\ \text{conserved residual.}\end{array} \tag{73}$$

The split into clustering matter and smooth acceleration is not a consequence of topology. It is obtained, within GR, only by combining homogeneous distances with perturbative gravitational observables. Compact topology is therefore kept as exact geometry, while the dark sector is reconstructed as an invariant residual and tested against public likelihoods. The strongest model-independent closure is the residual tensor itself; any further closure is a dynamical model choice constrained by perturbative observables.

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## DATA AND CODE AVAILABILITY

No proprietary data are used. The paper defines an analytic and statistical reconstruction pipeline. A numerical implementation should use public CMB, BAO, supernova, lensing, growth, cluster, and galaxy-dynamics likelihoods or data products with stated priors.

## AUTHOR CONTRIBUTIONS

Andrei Preece and Boris Batenin contributed to the conceptual formulation, invariant reconstruction framework, proof structure, and manuscript preparation.

### Appendix A: Pressure reconstruction

From Eq. (24),

$$\epsilon_{\mathrm{tot}} = \epsilon_{\mathrm{crit},0}\left(E^2-\Omega_K a^{-2}\right).$$

Therefore

$$\frac{1}{\epsilon_{\mathrm{crit},0}}\frac{\mathrm{d}\epsilon_{\mathrm{tot}}}{\mathrm{d}\ln a}=\frac{\mathrm{d}E^2}{\mathrm{d}\ln a}+2\Omega_K a^{-2}.$$

Using

$$p_{\mathrm{tot}}=-\epsilon_{\mathrm{tot}}-\frac{1}{3}\frac{\mathrm{d}\epsilon_{\mathrm{tot}}}{\mathrm{d}\ln a}$$

gives Eq. (26).

### Appendix B: Closed distances and the local flat-metric limit

The closed transverse comoving distance is

$$D_M = R_0\sin(D_C/R_0).$$

Since $\sin x = x - x^3/6 + O(x^5)$, the large-radius limit $R_0 \to \infty$ gives $D_M \to D_C$ with corrections of order $(D_C/R_0)^2 \sim |\Omega_K|$. This is a local geometric limit of the distance formula, not a topological identification of compact $S^3$ with $\mathbb{R}^3$.

### Appendix C: Flat-space limit of the invariant closure

The large-radius limit used in this paper is local. One sends

$$R_0 \to \infty, \qquad \Omega_K = -\frac{c^2}{H_0^2 R_0^2} \to 0,$$

while holding fixed physical distances such as $r = R_0\chi$, the line-of-sight comoving distance $D_C(z)$, and the continuum wavenumber

$$k = \frac{n+1}{R_0}.$$

This is not a topological identification of $S^3$ with $\mathbb{R}^3$; it is the local geometric and spectral limit on scales much smaller than $R_0$.

The global geometric invariants behave as

$$\begin{aligned} &\mathrm{Vol}(S^3_{R_0}) = 2\pi^2 R_0^3 \to \infty, \\ &{}^{(3)}\mathcal{R} = \frac{6}{R_0^2} \to 0, \qquad \mathrm{diam}(S^3_{R_0}) = \pi R_0 \to \infty. \end{aligned} \tag{C1}$$

For a ball of fixed physical radius $r = R_0\chi$,

$$V(\chi) = \frac{4\pi}{3} r^3 + O(r^5/R_0^2), \qquad A(\chi) = 4\pi r^2 + O(r^4/R_0^2).$$

The causal-accessibility fraction in Eq. (9) has a subtle but important limit. For finite propagation time $T$,

$$\Gamma(T) = \frac{\theta - \frac{1}{2}\sin 2\theta}{\pi} = \frac{2}{3\pi}\left(\frac{cT}{R_0}\right)^3 + O(R_0^{-5}) \to 0,$$

because the total spatial volume diverges. The local accessible volume itself, however, tends to the Euclidean value $4\pi(cT)^3/3$. Similarly, the antipodal condition in Eq. (10) is not automatically satisfied in the flat limit; rather, the antipode recedes to infinite distance and the condition ceases to be a finite-distance observable.

The distance map reduces continuously to the flat-FLRW one:

$$\begin{aligned} D_M(z) &= R_0 \sin\!\left(\frac{D_C(z)}{R_0}\right) \\ &= D_C(z)\left[1 - \frac{1}{6}\left(\frac{D_C(z)}{R_0}\right)^2 + O\!\left(\frac{D_C^4}{R_0^4}\right)\right], \end{aligned} \tag{C2}$$

so that

$$D_M \to D_C, \qquad D_A \to \frac{D_C}{1+z}, \qquad D_L \to (1+z)D_C,$$

with relative corrections

$$\left(\frac{D_C}{R_0}\right)^2 = (-\Omega_K)\left(\frac{H_0 D_C}{c}\right)^2.$$

The BAO combinations therefore become

$$D_V \to \left[z D_C^2 D_H\right]^{1/3}, \qquad F_{\mathrm{AP}} \to \frac{D_C H}{c},$$

while Etherington reciprocity is unchanged.

The background reconstruction reduces to the standard flat residual:

$$\frac{\epsilon_{\mathrm{tot}}}{\epsilon_{\mathrm{crit},0}} = E^2 - \Omega_K a^{-2} \to E^2, \tag{C3}$$

$$\begin{aligned} \frac{p_{\mathrm{tot}}}{\epsilon_{\mathrm{crit},0}} &= -\frac{1}{3}\left[\frac{\mathrm{d}E^2}{\mathrm{d}\ln a} + 3E^2 - \Omega_K a^{-2}\right] \\ &\to -\frac{1}{3}\left[\frac{\mathrm{d}E^2}{\mathrm{d}\ln a} + 3E^2\right], \end{aligned} \tag{C4}$$

and therefore

$$\frac{\epsilon_{\mathcal{D}}}{\epsilon_{\mathrm{crit},0}} \to E^2 - \Omega_{\mathrm{b}} a^{-3} - \Omega_\gamma a^{-4} - \Omega_\nu(a),$$

with the corresponding flat expression for $p_{\mathcal{D}}$ obtained by subtracting the visible pressure. The acceleration scalar, the cold-clustering split, the smooth residual, and the filters of Sec. X reduce with the same substitution $\Omega_K \to 0$.

The scalar, vector, and tensor spectra also have a continuous local limit. With $k = (n+1)/R_0$,

$$\frac{n(n+2)}{R_0^2} = k^2 - \frac{1}{R_0^2} \to k^2, \tag{C5}$$

$$\frac{n(n+2)-1}{R_0^2} = k^2 - \frac{2}{R_0^2} \to k^2, \tag{C6}$$

$$\frac{n(n+2)-2}{R_0^2} = k^2 - \frac{3}{R_0^2} \to k^2. \tag{C7}$$

For scalar modes the compact replacement rule becomes

$$\begin{aligned} &\frac{1}{\mathrm{Vol}(S^3_{R_0})}\sum_{n=0}^{\infty}(n+1)^2 F\!\left(\frac{n+1}{R_0}\right) \\ &\quad\longrightarrow \frac{1}{2\pi^2}\int_0^\infty k^2 F(k)\,\mathrm{d}k = \int \frac{\mathrm{d}^3k}{(2\pi)^3} F(k). \end{aligned} \tag{C8}$$

with the vector and tensor sectors giving the corresponding continuum polarization sums. Equivalently, the compact spectral covariance tends to the flat Fourier covariance after the sum over hyperspherical shells is converted to the integral over $k$; the familiar angular kernel $\sin(kr)/(kr)$ appears inside the limiting integral,

$$C(r) \to \frac{1}{2\pi^2}\int_0^\infty k^2 P(k)\frac{\sin(kr)}{kr}\,\mathrm{d}k.$$

Thus Eq. (22) becomes the usual flat line-of-sight expression, up to the same normalization convention for transfer

functions and primordial spectra used in the closed calculation.

Finally, the invariant closure itself is continuous. The order-by-order tensor identity, the complete target $\mathcal{C}_{\rm inv}$, the posterior factorization, and the dynamical filters do not acquire singular terms as $\Omega_K \to 0$. Compact $S^3$ therefore supplies a finite closed infrared bookkeeping system whose local observables and residual identities reduce smoothly to the standard spatially flat FLRW framework.

### Appendix D: Requirements for a physical explanation

A proposed physical origin for the residual is not part of this formalism until it supplies:

1. an action, modified field equation, or stress tensor;
2. conservation laws or an explicit interaction term;
3. background functions $\epsilon(a)$ and $p(a)$;
4. perturbation equations, sound speed, and anisotropic stress;
5. lensing and growth predictions;
6. consistency with BBN, CMB, BAO, supernovae, clusters, and galaxies;
7. a local flat-metric and closed-geometry comparison;
8. public likelihood results against ΛCDM, curved ΛCDM, $w$CDM, modified gravity, and nonparametric dark-energy reconstructions.


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